GRADE 10 · TERM 1

Think, Plan, Code & Count in Binary

A pencil-and-paper workbook: think it through first, learn the Delphi, then prove you can do it.

Name: _____

Class: _____ Date: _____

How to use this workbook

1. Work with a **pencil**. Mistakes are how we learn: rub out and try again.
2. Start each unit with **Think First**. Try it on your own before you read the explanation.
3. Read **Learn It**, then do the activities. Trace tables and code are written **by hand**.
4. At the end of each unit, tick the "I can" boxes you are sure of. Write down anything you still want to ask.
5. The **checkpoint test** is done under test conditions, with no notes.

In this workbook

1. Plan before you code
2. Meet Delphi
3. Boxes with labels
4. Calculating
5. Making a choice
6. Bits and bytes on paper
7. Term checkpoint test

> **Matches the Annual Teaching Plan:** Term 1 — algorithms and problem solving; the Delphi IDE; variables, data types, input and output; operators, Random, Round and Sqrt; IF and IF…ELSE; number systems, ASCII and storage units.

UNIT 1 Plan before you code

Good programmers do not start typing. They plan on paper first. In this unit you will learn the planning tools: IPO tables, flowcharts, pseudocode and trace tables.

By the end of this unit I can: (tick each box when you are sure)

- ☐ explain what an algorithm is and what makes one good
- ☐ complete an IPO table
- ☐ read and draw a simple flowchart
- ☐ write pseudocode
- ☐ test an algorithm with a trace table
- ☐ compare two algorithms

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

1.1 Tea for a robot

A robot has arrived at your house. It does **exactly** what you tell it, nothing more, and it has never seen a kettle. Write numbered steps so that it makes **one cup of rooibos tea with milk**.
Now swap with a partner. They play the robot and follow your steps *literally*. Circle every step where the robot got stuck or did something silly.

LEARN IT

Algorithms and the problem-solving cycle

An **algorithm** is a clear, step-by-step list of instructions that solves a problem. A recipe is an algorithm. A good algorithm is **clear** (each step has only one meaning), **in the right order**, **finite** (it ends), **correct** and **efficient** (no wasted steps).
Programmers follow a cycle: **1** Understand the problem → **2** Plan (IPO table, flowchart, pseudocode) → **3** Write the code → **4** Test (trace table) → **5** Fix the errors and test again.

LEARN IT

IPO tables

Before you plan the steps, list what the problem needs. An **IPO table** has three columns: **I**nput (the data you are given), **P**rocess (what you do with it, written as a formula or a decision, not just "calculate") and **O**utput (what you show at the end).

Problem	Input	Process	Output
Total money from a taxi	passengers, fare	total = passengers × fare	total

1.2 Complete the IPO tables [6]

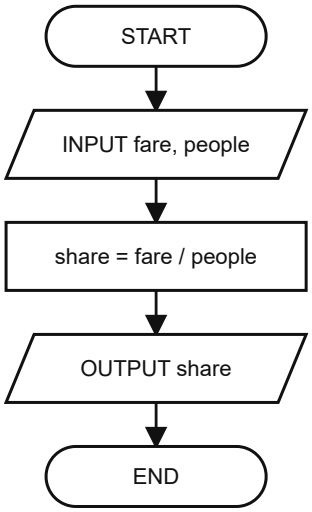
Fill in the empty cells. Write the Process as a formula.

Problem	Input	Process	Output
Area of a netball court			
Cost of loaves at R17 each			
Average of three test marks			

LEARN IT

Flowcharts and pseudocode

A **flowchart** draws an algorithm and arrows show the direction. **Pseudocode** writes it in plain, structured English. Neither is real Delphi. Here is the same algorithm both ways.



- **Oval**: start or end
- **Parallelogram**: input or output
- **Rectangle**: a process (calculation)
- **Diamond**: a yes/no decision

Pseudocode

```
START
  INPUT fare, people
  SET share TO fare / people
  OUTPUT share
END
```

Pseudocode rules: key words in CAPITALS, indent inside IF, one action per line, begin with START and end with END.

1.3 Fill in the blanks [5]

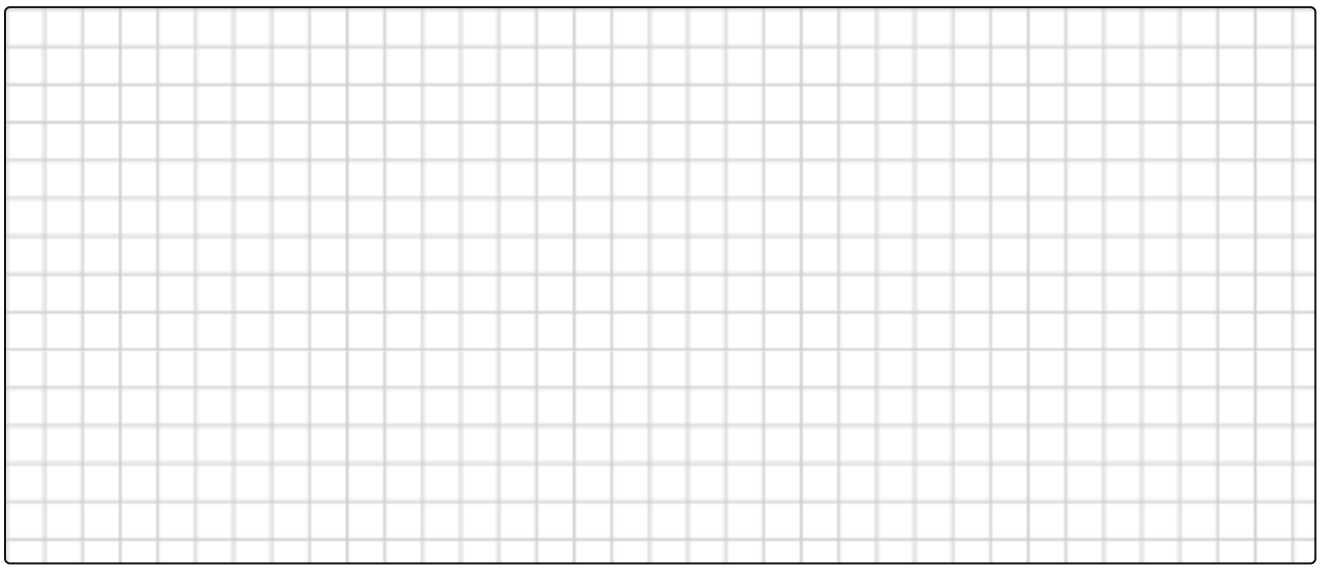
The shape that marks the start and the end of a flowchart is an _____. A _____ shows input or output, a rectangle shows a _____ (a calculation) and a _____ shows a decision. An _____ table lists the Input, Process and Output.

1.4 Draw the flowchart [6]

A learner buys some pies. Draw the flowchart for this pseudocode. Use the correct shapes and arrows.

Pseudocode

```
START
  INPUT price, number, paid
  SET total TO price * number
  SET change TO paid - total
  OUTPUT change
END
```



1.5 Write the pseudocode [4]

The school raises money with tickets at **R35** each. The expenses are **R400**. Write pseudocode that reads the number of tickets sold and outputs the profit.

1.6 Trace the algorithm [6]

A shop gives free delivery on orders of R200 or more. Otherwise delivery costs R25. Complete the trace table for each test value.

Pseudocode

```
INPUT total
IF total >= 200 THEN
  SET fee TO 0
ELSE
  SET fee TO 25
ENDIF
SET pay TO total + fee
OUTPUT pay
```

Test value (total)	total >= 200 ?	fee	pay (output)
150			
200			
320			
199			

1.7 Compare the algorithms [5]

All four algorithms try to find the **average of three marks**. Read them, then answer.

Algorithm P

```
INPUT m1, m2, m3
SET average TO total / 3
SET total TO m1 + m2 + m3
OUTPUT average
```

Algorithm Q

```
INPUT m1, m2, m3
SET total TO m1 + m2 + m3
SET average TO total / 3
OUTPUT average
```

Algorithm R

```
INPUT m1, m2, m3
SET total TO m1 + m2 + m3
SET total2 TO m1 + m2 + m3
SET average TO total / 3
OUTPUT average
```

Algorithm S

```
Get the marks
Add them up and do the sums
Show the answer
```

1	The steps are in the **wrong order**.	**P** **Q** **R** **S**
2	A step is **too vague** for a computer to follow.	**P** **Q** **R** **S**
3	Correct, with **no wasted steps**.	**P** **Q** **R** **S**
4	**Repeats** a calculation it has already done.	**P** **Q** **R** **S**
5	Which two give the **right answer**?	**P+Q** **Q+R** **R+S**

UNIT 2 Meet Delphi

Delphi lets you design a window by dragging components onto a form and then write code that brings it to life. First, learn your way around the workshop.

By the end of this unit I can: (tick each box when you are sure)

- ☐ name the main parts of the Delphi IDE and say what each is for
- ☐ explain class, object, component, property, method and event
- ☐ use the correct prefix for a component name
- ☐ change properties in code with Component.Property := Value;
- ☐ plan a program with a TOE chart

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

2.1 The vending machine

A vending machine sits there doing **nothing** until something happens. Describe two things a customer can do to it, and what the machine does in response. Then answer: what would the machine do if nobody touched it?

_____

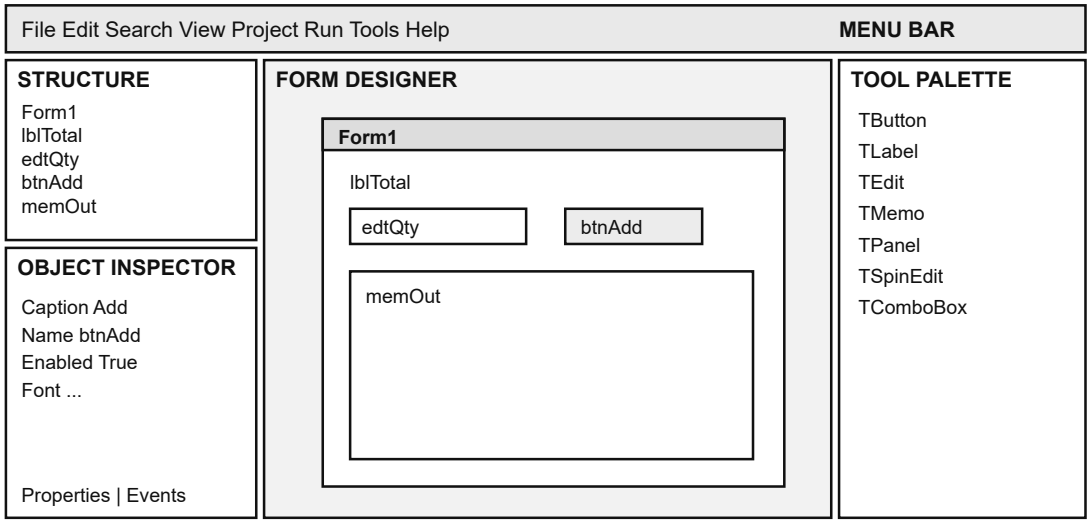
_____

_____

LEARN IT

The Delphi IDE

The **IDE** (Integrated Development Environment) is your workshop: an editor, a compiler and a debugger in one program.



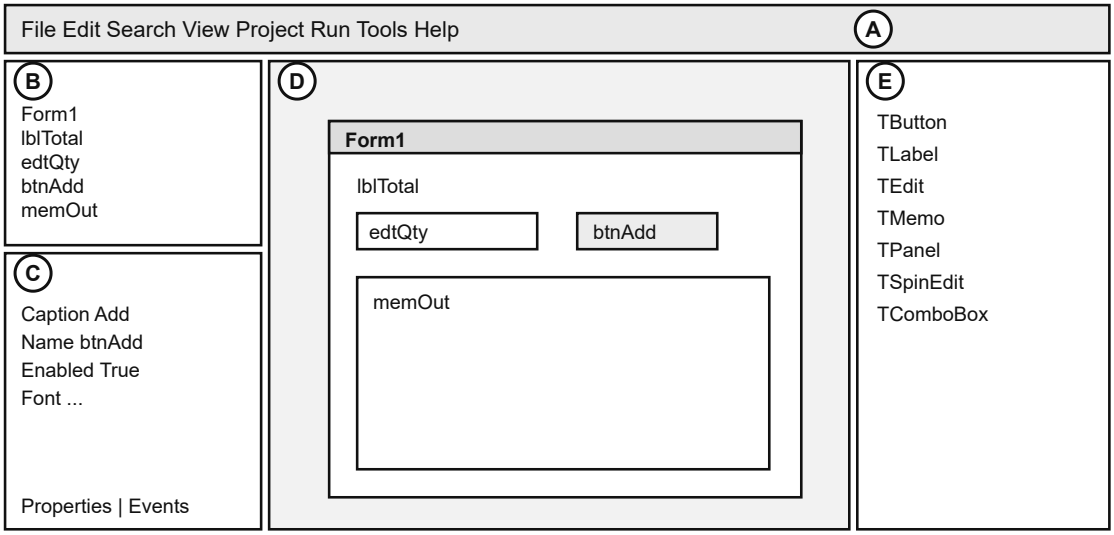
- **Menu bar**: Save All, Run, and so on
- **Tool Palette**: components to click and place on the form
- **Form Designer**: the window you are building
- **Object Inspector**: properties and events of the selected component
- **Structure panel**: a tree of every component

Press **F12** to flip between the form and the code editor. Press **F9** to compile and run. Save each project in **its own folder** (unit as *_u.pas*, project as *_p.dpr*).

2.2 Label the IDE

[5]

Write the name of each part. The circled letters are on the picture.



Letter	Name of the part
A	
B	
C	
D	
E	

2.3 Which part of the IDE?

[4]

Circle the part you would use.

1	Drag a new button onto the form.	**Tool Palette** **Object Inspector** **Menu bar**
2	Change the Caption of a label.	**Tool Palette** **Object Inspector** **Menu bar**
3	See a list of every component on the form.	**Structure panel** **Menu bar** **Tool Palette**
4	Save All, or run the program.	**Object Inspector** **Menu bar** **Form Designer**

LEARN IT

Classes, objects, components and events

A **class** is the plan, for example *TButton*. An **object** is a real thing built from the plan, for example *btnPay*. A **component** is an object you place on a form. Every component has **properties** (how it looks, such as Caption), **methods** (what it can do, such as Hide) and **events** (what can happen to it, such as OnClick). Name components with a **prefix** that shows the type: **btn** button, **lbl** label, **edt** edit box, **mem** memo, **sed** spin edit, **cmb** combo box. A label shows its text in **Caption**; an edit box keeps its text in **Text**. Delphi code runs only inside an **event handler**, in this format:

```Delphi
procedure TForm1.btnGreetClick(Sender: TObject);
begin
  lblWelcome.Caption := 'Welcome to the spaza shop';   // Component.Property := Value;
  btnGreet.Caption := 'Thanks';
  lblWelcome.Font.Color := clGreen;   // spelt Color, not Colour
  edtName.Clear;                      // a method
  memOut.Hide;                        // another method
end;
```

2.4 Prefixes and names [8]

Write the prefix, then a good name (prefix + meaning) for the component described.

Component	Prefix	A good name for...
TButton		
TLabel		
TEdit		
TMemo		

2.5 Debug: component statements [5]

Each line has one error. Write the correct line.

```Delphi
lblResult.Caption = 'Total';
'Welcome' := lblMsg.Caption;
btnOk.Caption := Click me;
edtName.Text := 'Thabo'
lblMsg.Font.Colour := clBlue;
```

2.6 Write the event handler [4]

When **btnReset** is clicked it must: set **lblTotal** to *R0.00*; clear **edtQty**; change its own caption to *Cleared*; and hide **memReceipt**.

```Delphi
procedure TForm1.btnResetClick(Sender: TObject);
```

2.7 Plan with a TOE chart [5]

A **TOE chart** lists each **T**ask, the **O**bject that does it and the **E**vent that starts it. A memo that only displays results has the event “None”. Complete the chart for a till program.

Task	Object	Event
Work out the total		
Clear the screen for the next customer		
Show the receipt lines		

UNIT 3 Boxes with labels

A program must remember things while it runs. It keeps them in variables: labelled boxes in memory. In this unit you learn to declare variables, choose the right type, and move data in and out.

By the end of this unit I can: (tick each box when you are sure)

- ☐ explain what a variable is
- ☐ choose the correct data type
- ☐ declare variables with correct prefixes
- ☐ use := to put a value in a variable
- ☐ convert between text and numbers
- ☐ get input and show output in Delphi

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

3.1 Swap the glasses

Glass A is full of orange juice and glass B is full of water. You must swap them so that A has water and B has juice. You have **one extra empty glass** and you may only pour from one glass into another. Write the steps.

_____

_____

_____

LEARN IT

Variables and data types

A **variable** is a named place in memory that holds **one value** that can change. Its **data type** decides what may go in. Declare it in a **var** section: *name : type;* Names start with a letter, have no spaces, and begin with a **prefix** for the type. Put a value in with the assignment operator **:=** (the value on the right goes into the box on the left).

Type	Holds	Example value	Prefix
Integer	whole numbers	12, -5, 1250	iCount
Real	numbers with decimals	12.50, 3.14	rPrice
String	text, in single quotes	'Thandi'	sName
Char	one character	'B'	cGrade
Boolean	True or False	True	bPaid

Delphi

```
var
  iCount : Integer;
  sName : string;
begin
  iCount := 12;
  sName := 'Thandi';
  iCount := iCount + 1;   // now 13: the old 12 is replaced
end;
```

3.2 Choose the type [10]

Choose the best data type for each item and write a variable name with the correct prefix.

Item to store	Data type	Variable name
Number of learners in a class		
Price of a loaf: R12.50		
A learner's surname		
The class letter, for example 'C'		
Has the fee been paid? (yes or no)		

3.3 Be the computer: assignment [5]

Write the value held by each variable **after** each line runs. Write ? if the variable has no value yet.

```Delphi
iApples := 5;
iOranges := 3;
iFruit := iApples + iOranges;
iApples := iApples - 2;
iFruit := iFruit * 2;
```

After line	iApples	iOranges	iFruit
1			
2			
3			
4			
5			

3.4 Be the computer: swap [3]

Start with iA = 8 and iB = 15. Show the values after each line.

```Delphi
iTemp := iA;
iA := iB;
iB := iTemp;
```

After line	iA	iB	iTemp
1			
2			
3			

LEARN IT

Input and output in Delphi

An edit box keeps what the user typed as **text**, and so does **InputBox** (a pop-up that asks a question). You must **convert** text to a number before you calculate, and a number back to text before you display it. **ShowMessage** pops up a message, **Caption** sets a label, **memOut.Lines.Add** adds a line to a memo, and **+** joins text together.

Function	Converts	Example
StrToInt	text → Integer	iAge := StrToInt(edtAge.Text);
StrToFloat	text → Real	rPrice := StrToFloat(edtPrice.Text);
IntToStr	Integer → text	lblOut.Caption := IntToStr(iTotal);
FloatToStr	Real → text	lblOut.Caption := FloatToStr(rAvg);

For money, **FloatToStrF(value, ffFixed, 8, 2)** shows exactly two decimals.

Delphi

```
var
  iFirst, iSecond, iSum : Integer;
  sName : string;
begin
  sName := InputBox('Name', 'Your name?', '');
  iFirst := StrToInt(edtFirst.Text);
  iSecond := StrToInt(InputBox('Second', 'Second number:', ''));
  iSum := iFirst + iSecond;
  lblAnswer.Caption := sName + ', the sum is ' + IntToStr(iSum);
  ShowMessage('Done!');
end;
```

3.5 Pick the conversion [4]

Circle the function that completes each statement.

1	iQty := _____(edtQty.Text);	**StrToInt** **IntToStr** **StrToFloat**
2	lblTotal.Caption := _____(iTotal);	**StrToInt** **IntToStr** **FloatToStr**
3	rPrice := _____(InputBox('Price', 'Price in rand:', ''));	**StrToInt** **StrToFloat** **FloatToStr**
4	lblAvg.Caption := _____(rAverage);	**IntToStr** **StrToFloat** **FloatToStr**

3.6 Debug: the till button [4]

Find **four** errors in this event handler and write the corrections.

Delphi

```
procedure TForm1.btnTotalClick(Sender: TObject);
var
  iQty : Integer;
  rPrice, rTotal : Real;
begin
  iQty := edtQty.Text;
  rPrice := 12.50;
  rTotal := iQty * rPrice
  lblTotal.Caption := rTotal;
  ShowMessage('Total: ' + rTotal);
end;
```

3.7 Write it: the till [8]

Write the code for **btnTotalClick**. The number of items is typed in **edtQty**. The price (with cents) is asked with an **InputBox**. Calculate the total and show it in **lblTotal** with an R in front and two decimals. Include the variable declarations.

UNIT 4 Calculating

Computers are fast calculators, but they follow strict rules about the order of work and about whole numbers and decimals. Learn the operators, then three useful functions: Random, Round and Sqrt.

By the end of this unit I can: (tick each box when you are sure)

- ☐ use + - * / DIV and MOD
- ☐ explain the difference between / and DIV
- ☐ apply the order of operations
- ☐ use Random, Round and Sqrt correctly
- ☐ choose Integer or Real for a result

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

4.1 Boxes and leftovers

(a) The tuck shop has **95 samoosas** to pack in boxes of **8**. How many boxes are completely full? How many samoosas are left over? How many boxes are needed if **every** samoosa must be packed?
(b) A day of load shedding adds up to **245 minutes**. How many hours and minutes is that?

LEARN IT

The operators

Delphi has six arithmetic operators. **/** always gives a **Real**, even when the answer looks whole. **DIV** gives the whole-number part of a division and **MOD** gives the remainder. Use **n MOD 2 = 0** to test for an even number and **n MOD 10** to get the last digit. **Inc(x)** adds 1 to x and **Dec(x)** subtracts 1.

Operator	Meaning	Example	Result
+	add	12 + 5	17
-	subtract	12 - 5	7
*	multiply	12 * 5	60
/	divide (a Real answer)	17 / 5	3.4
DIV	whole-number division	17 DIV 5	3
MOD	remainder	17 MOD 5	2

LEARN IT

Order of operations

Delphi works in this order: **1** brackets ( ), **2** then * / DIV MOD, left to right, **3** then + and -, left to right. Use brackets to make your meaning clear. Example: **7 + 15 MOD 4 * 2**. MOD and * come first, from the left: 15 MOD 4 = 3, then 3 * 2 = 6, then 7 + 6 = 13. To average three marks you must write **(iMark1 + iMark2 + iMark3) / 3**: without the brackets only iMark3 would be divided.

4.2 DIV and MOD

[10]

Work out each answer. Show your thinking in the margin if you need to.

Calculation	DIV	MOD
29 ÷ 4		
100 ÷ 7		
45 ÷ 9		
8 ÷ 12		
1234 ÷ 10		

4.3 Order of operations

[6]

Work out each answer. Remember that / gives a Real.

Expression	Answer
6 + 4 * 3	
(6 + 4) * 3	
20 - 12 DIV 4	
17 MOD 5 + 2 * 3	
100 DIV (3 + 2) * 4	
10 / 4	

4.4 Integer or Real?

[6]

Can the answer be stored in an **Integer** variable, or is a **Real** variable needed?

1	17 DIV 5	**Integer**	**Real**
2	17 / 5	**Integer**	**Real**
3	17 MOD 5	**Integer**	**Real**
4	12 * 3	**Integer**	**Real**
5	20 / 4 (the answer looks whole)	**Integer**	**Real**
6	Sqrt(49)	**Integer**	**Real**

LEARN IT

Random, Round and Sqrt

Random(n) gives a whole number from **0 to n-1**, so add to shift it: *Random(6) + 1* is a die (1 to 6). Call **Randomize** once when the program starts. **Round(x)** gives the nearest whole number. A value exactly halfway goes to the **even** neighbour, so Round(2.5) is 2 but Round(3.5) is 4. **Sqrt(x)** gives the square root as a Real. It cannot take a negative number.

Delphi

```
Randomize;                        // once, at the start
iDie := Random(6) + 1;            // 1, 2, 3, 4, 5 or 6
iNear := Round(12.678);           // 13
rRoot := Sqrt(49);                // 7.0 (a Real)
lblSide.Caption := IntToStr(Round(Sqrt(30)));   // 5
```

4.5 Fill in the blanks [10]

Round(6.3) = _____. Round(6.7) = _____. Round(2.5) = _____. Round(3.5) = _____. Sqrt(64) = _____. Round(Sqrt(30)) = _____.

To get a random number from 5 to 15 write Random(_____) + _____. Random(20) + 10 gives a number from _____ to _____.

4.6 Be the computer: DIV, MOD and Round [5]

Complete the table for the code.

Delphi

```
iA := 50;
iB := 8;
iQuot := iA div iB;
iRem := iA mod iB;
rDiv := iA / iB;
iCheck := iQuot * iB + iRem;
iRound := Round(rDiv);
```

Variable	Value stored
iQuot	
iRem	
rDiv	
iCheck	
iRound	

4.7 Write it: packing [6]

The number of cupcakes is typed in **edtCakes**. Each box holds **12**. Show the number of full boxes in **lblBoxes** and the number left over in **lblLeft**. Include variable declarations.

4.8 Write it: two dice [4]

Write code that rolls **two dice** and shows their total in **lblTotal**.

4.9 Debug: three errors [3]

Find **three** errors. Each statement should give the result in its comment.

```Delphi
iAverage := iM1 + iM2 + iM3 / 3;   // average of three marks (iAverage is an Integer)
iDie := Random(6);                 // a die: 1 to 6
```

UNIT 5 Making a choice

Programs make decisions all the time: pass or fail, free delivery or not, a bonus or no bonus. You will learn to ask a True/False question in code and act on the answer.

By the end of this unit I can: (tick each box when you are sure)

- ☐ write a condition with =, <>, >, <, >= and <=
- ☐ write and trace IF…THEN and IF…THEN…ELSE
- ☐ use begin…end for more than one statement
- ☐ see how a flowchart becomes IF code, and draw a flowchart from code
- ☐ choose test values that include the boundary

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

5.1 Rules we live by

Write each rule in the form **IF … THEN … ELSE …**

(a) A taxi driver charges children half price. (b) Your school does not open when there is a strike. (c) You study for the test only if you did badly in the last one.

LEARN IT

Conditions and IF

A **condition** is a question with the answer True or False. Compare with: **=** equal, **<>** not equal, **>**, **<**, **>=** and **<=**. **IF…THEN** runs a statement only when the condition is True. **IF…THEN…ELSE** chooses one of two paths. Three rules: put **no semicolon before else**; use **begin…end** when a branch has more than one statement; indent the branches.

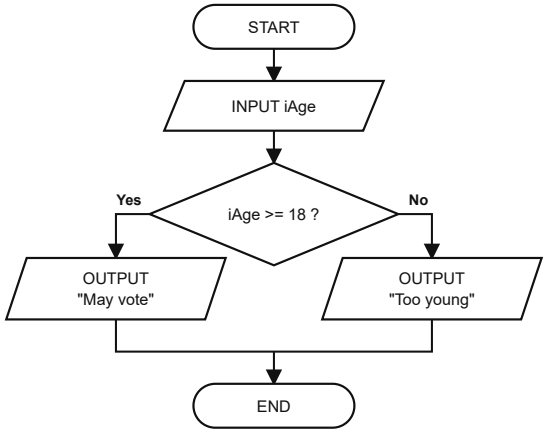
Delphi

```
if iMark >= 50 then
  lblResult.Caption := 'Pass';          // one-way IF
if iStock = 0 then
  lblNote.Caption := 'Sold out'         // no semicolon before else
else
  lblNote.Caption := 'In stock';
if iMark >= 50 then
begin                                   // two statements need begin...end
  lblResult.Caption := 'Pass';
  btnRetry.Visible := False;
end
else
  lblResult.Caption := 'Fail';
```

LEARN IT

From flowchart to code

A decision diamond becomes an IF. The Yes path is the THEN part and the No path is the ELSE part.



Delphi

```
iAge := StrToInt(edtAge.Text);
if iAge >= 18 then
  lblMsg.Caption := 'May vote'
else
  lblMsg.Caption := 'Too young';
```

5.2 Write the condition [5]

Write each sentence as a Delphi condition.

In words	Condition
The mark is at least 50	
The learner is younger than 14	
The number is even	
The name is Zinhle	
The price is not R10	

5.3 Trace the code [6]

Trace the code for each age. The standard fare is R20.

Delphi

```
rFare := 20;
if iAge < 13 then
  rFare := rFare / 2;
if iAge >= 60 then
  rFare := rFare - 5;
```

iAge	iAge < 13 ?	iAge >= 60 ?	rFare at the end
7			
13			
60			

5.4 Debug: the IFs [4]

Find **four** errors and write the corrections.

Delphi

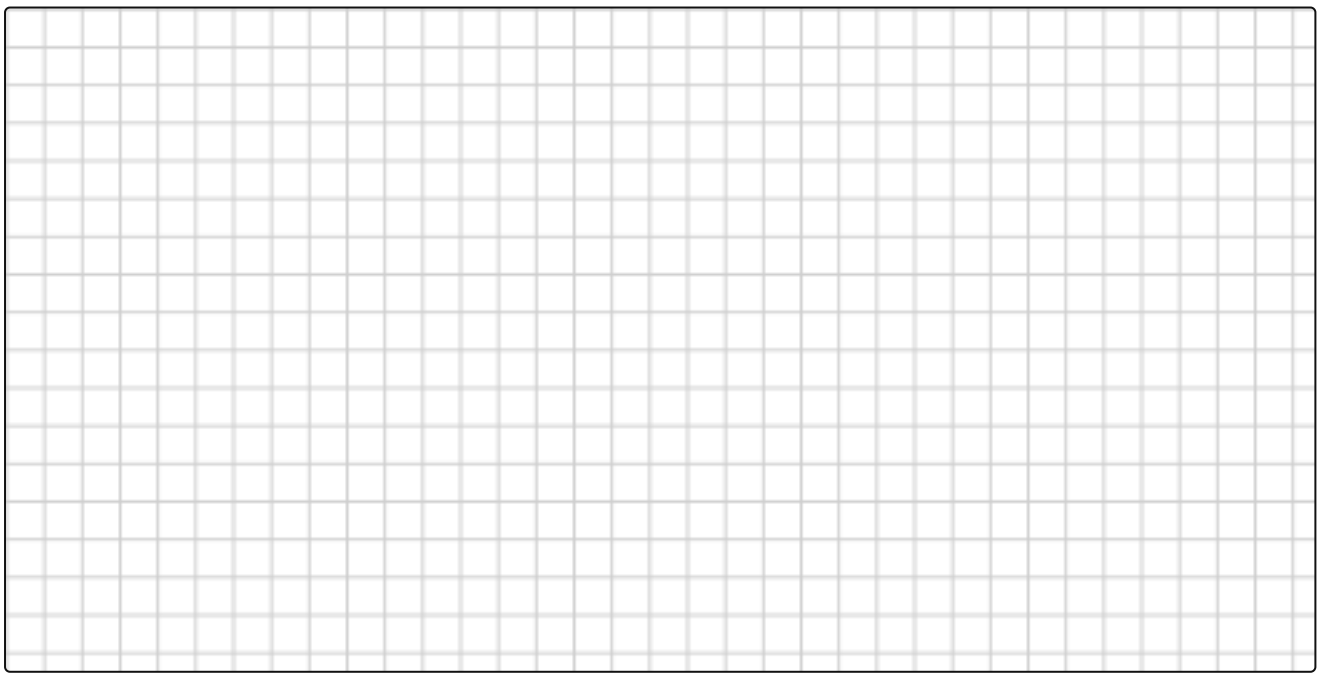
```
if iMark => 50 then
  lblResult.Caption := 'Pass';
else
  lblResult.Caption := 'Fail';
  btnRetry.Visible := True;
if sName = Thabo then
  ShowMessage('Hi Thabo');
```

5.5 Code to flowchart [6]

Draw the flowchart for this code.

Delphi

```
iSpeed := StrToInt(edtSpeed.Text);
if iSpeed > 60 then
  rFine := 10 * (iSpeed - 60)
else
  rFine := 0;
lblFine.Caption := FloatToStr(rFine);
```



5.6 Write it: delivery [6]

The order total (with cents) is typed in **edtTotal**. Delivery is **R0** for orders of **R200 or more**, otherwise **R25**. Show the amount to pay in **lblPay**.

5.7 Write it: even or odd [4]

A number is typed in an InputBox. Show *Even* or *Odd* in **lblKind**.

5.8 Choose the test values [4]

A shop gives a discount when the quantity is **10 or more**. Write three test values for the quantity that test the rule well, and say what you expect for each.

UNIT 6 Bits and bytes on paper

Everything inside a computer is stored as 1s and 0s. Here you work with binary, decimal and hexadecimal by hand, see how letters become numbers, and measure how much space data needs.

By the end of this unit I can: (tick each box when you are sure)

- ☐ explain why computers use binary
- ☐ convert between binary and decimal
- ☐ convert between hexadecimal and binary or decimal
- ☐ give the ASCII code of a letter and say what Unicode adds
- ☐ convert between bytes, KB, MB, GB and TB
- ☐ tell data from information

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

6.1 Light switches

A computer is built from billions of tiny switches, each **ON** or **OFF**. Think of a row of light switches. How many different patterns can you make?

1 switch gives _____ patterns. 2 switches give _____. 3 switches give _____. 4 switches give _____. 8 switches give _____.

Spot the rule and explain it in one sentence.

_____

_____

LEARN IT

Binary place values

Decimal uses 10 digits and every column is worth 10 times the one on its right. **Binary** (base 2) uses only **0 and 1** and every column is worth **2 times** the one on its right. A **bit** is one binary digit; a **byte** is 8 bits.

128	64	32	16	8	4	2	1
0	0	1	0	1	1	0	1

Binary to decimal: add the place values where there is a 1. 00101101 = 32 + 8 + 4 + 1 = **45**.

Decimal to binary: go from the biggest place value down. Does it fit? Write 1 and subtract; if not, write 0. For 77: 64 fits (13 left); 32 no; 16 no; 8 fits (5 left); 4 fits (1 left); 2 no; 1 fits. So 77 = 01001101.

6.2 Binary and decimal [8]

Convert. Write decimal answers as ordinary numbers and binary answers as 8 bits.

Start with	Convert to	Answer
1011 (binary)	decimal	
10010 (binary)	decimal	
00101101 (binary)	decimal	
10000001 (binary)	decimal	
6	8-bit binary	
19	8-bit binary	
37	8-bit binary	
200	8-bit binary	

LEARN IT

Hexadecimal

Long binary numbers are hard to read. **Hexadecimal** (base 16) has 16 digits: 0 to 9 and then **A=10, B=11, C=12, D=13, E=14, F=15**. One hex digit is exactly **4 bits**, so you can convert in groups of four bits. Hex is used for colour codes (#FF0000 is red) and memory addresses.

Binary	0101	1001	1010	1100	1111
Hex	5	9	A	C	F

Binary to hex: split into groups of 4: 1011 0111 → B and 7 → **B7**.
Hex to decimal: each column is worth 16 times the one on its right: 2A = 2 × 16 + 10 = **42**.
Decimal to hex: divide by 16: 200 ÷ 16 = 12 remainder 8, so 200 = C8.

6.3 Hexadecimal conversions [6]

Convert.

Start with	Convert to	Answer
10110111 (binary)	hex	
7E (hex)	8-bit binary	
2F (hex)	decimal	
FF (hex)	decimal	
100	hex	
255	hex	

LEARN IT

Text as numbers: ASCII and Unicode

A computer stores a letter as a number. **ASCII** gives each character a code that is stored as one byte: space = 32, digits **0 to 9 = 48 to 57**, capital letters **A to Z = 65 to 90**, small letters **a to z = 97 to 122**. A small letter is always **32 more** than its capital. ASCII covers only English. **Unicode** (for example UTF-8) covers more than 140 000 characters, including isiZulu and isiXhosa letters, Arabic, Chinese and emoji.

A	B	Z	a	z	0	9	space
65	66	90	97	122	48	57	32

6.4 ASCII [8]

Count along from the codes in the table above to find each ASCII code (decimal), then write its 8-bit binary code.

Character	ASCII code	8-bit binary
C		
H		
d		
7		

6.5 Decode the message [2]

Each number is the ASCII code of a capital letter. What word is hidden in **83 65 70 69**?

LEARN IT

Measuring storage

A bit is one 0 or 1. A **byte** is 8 bits and stores one ASCII character. Computers count in powers of two, so each unit below is **1 024** times the one before.

Unit	Size	Example
1 byte (B)	8 bits	one letter
1 kilobyte (KB)	1 024 bytes	half a page of text
1 megabyte (MB)	1 024 KB	a photo, a song
1 gigabyte (GB)	1 024 MB	a film, a game
1 terabyte (TB)	1 024 GB	a laptop hard drive

To go to a **smaller** unit multiply by 1 024; to go to a **bigger** unit divide by 1 024.

6.6 Storage conversions [5]

Convert. Use 1 024 each time.

Question	Answer
2 KB = ? bytes	
3 MB = ? KB	
5 GB = ? MB	
4096 KB = ? MB	
How many 4 MB songs fit on a 2 GB flash drive?	

6.7 Data or information? [6]

Data is raw facts with no meaning on their own. **Information** is data that has been processed so that it means something.

1	12, 7, 15, 9, 20	**Data**	**Information**
2	Grade 10A has the highest average: 68%	**Data**	**Information**
3	Thabo 54, Aisha 71, Pieter 49	**Data**	**Information**
4	A graph showing which taxi rank is busiest on Fridays	**Data**	**Information**
5	45.00, 12.50, 99.99 from till slips	**Data**	**Information**
6	Total sales this week: R2 340	**Data**	**Information**

TERM CHECKPOINT — Algorithms, Delphi basics and number systems

Time: 60 minutes **Total:** 49 marks **Mark:** _____ / 49

No notes, no computer. Show all your working. Write Delphi code neatly and use correct indentation.

SECTION A — Trace and predict [12]

A1 [5]

Complete the table for the code.

```Delphi
iA := 23;
iB := 5;
iC := iA div iB;
iD := iA mod iB;
iE := iC * 3 + iD;
rF := iA / iB;
iA := iA - iE;
```

Variable	Value stored
iC	
iD	
iE	
rF	
iA (at the end)	

A2 [3]

Write exactly what is displayed.

```Delphi
iPrice := 80;
iQty := 3;
iTotal := iPrice * iQty;
if iTotal > 200 then
  iTotal := iTotal - 20
else
  iTotal := iTotal + 15;
ShowMessage('Total: R' + IntToStr(iTotal));
```

A3 **[4]**

(a) What is the value of iNext? (b) What are the smallest and largest values iRoll can have?

Delphi

```
rNum := Sqrt(81);
iNext := Round(rNum) + 5 mod 3;
iRoll := Random(5) + 2;
```

SECTION B — Algorithms and flowcharts [13]

B1 **[3]**

Write an IPO table (Input, Process, Output) for this problem: *the cost of loaves at R15 each plus a R5 delivery fee*. Write the Process as a formula.

B2 **[5]**

Trace the algorithm for each test value.

Pseudocode

```
INPUT qty
SET price TO qty * 12
IF qty >= 10 THEN
  SET price TO price - 10
ENDIF
OUTPUT price
```

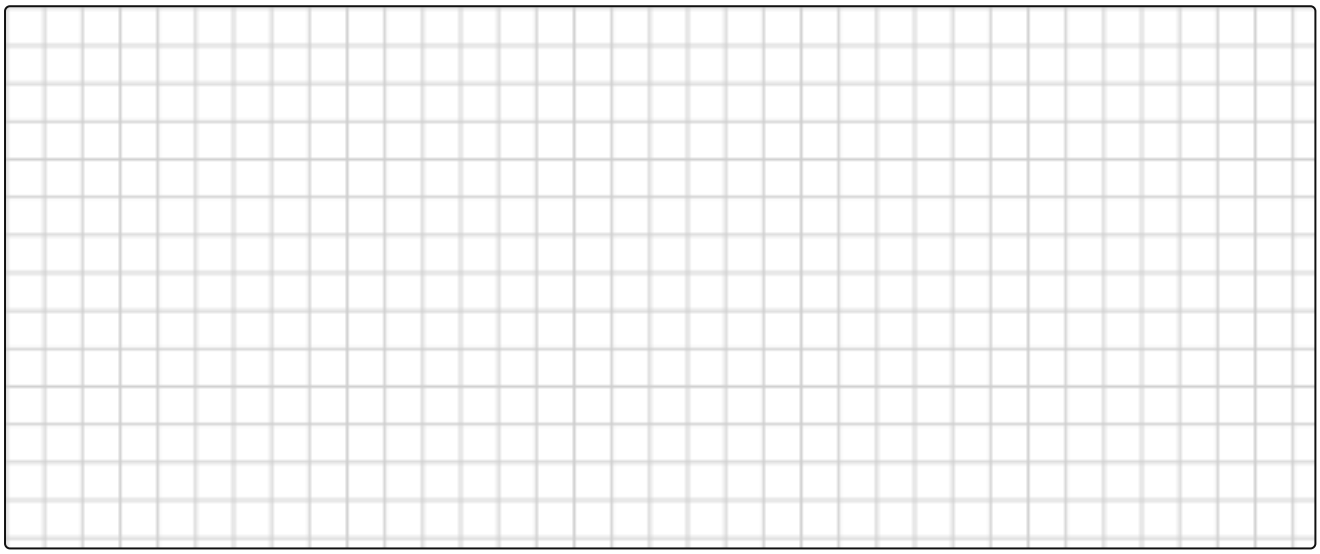
qty	qty >= 10 ?	price (output)
4		
10		
15		

B3 **[5]**

Draw the flowchart for this algorithm.

Pseudocode

```
START
  INPUT mark
  IF mark >= 50 THEN
    OUTPUT "Pass"
  ELSE
    OUTPUT "Fail"
  ENDIF
END
```



SECTION C — Write the Delphi code [15]

C1 **[3]**

Find **three** errors in this code and write the corrections.

Delphi

```
procedure TForm1.btnCalcClick(Sender: TObject);
var
  iPrice, iQty : Integer;
  rTotal : Real;
begin
  iPrice := StrToInt(edtPrice.Text);
  iQty := edtQty.Text;
  rTotal := iPrice * iQty * 1.15
  lblTotal.Caption := rTotal;
end;
```

C2 **[6]**

An InputBox asks for the number of minutes of load shedding (a whole number). Display it in **lblTime** as hours and minutes, for example *3 h 10 min* for 190 minutes. Include declarations.

C3 **[6]**

A game rolls **one die**. If the number is a **6**, display *Bonus!* in **lblMsg**. Otherwise display *You rolled a* followed by the number.

SECTION D — Number systems and data [9]

D1 **[5]**

Convert.

Start with	Convert to	Answer
10110 (binary)	decimal	
1101001 (binary)	decimal	
37	8-bit binary	
10101111 (binary)	hex	
2C (hex)	decimal	

D2 **[4]**

A byte is made of _____ bits. 3 KB = _____ bytes. If A is 65 in ASCII, then C is _____. 4 GB = _____ MB.

COMING UP NEXT TERM · warm-up

Two rules at once

Next term your programs will ask **two questions at the same time**. Try it first with ordinary school rules.

Rule 1: A learner may enter the 400 m race if she is in Grade 10 **AND** she has handed in her permission letter.
Rule 2: A learner gets free entry to the athletics evening if she is in the choir **OR** in the athletics team.

Learner	Grade	Letter handed in?	Choir?	Athletics team?	Rule 1	Rule 2
Thandi	10	Yes	No	Yes		
Aisha	10	No	Yes	No		
Pieter	9	Yes	No	No		

In your own words: when is a rule with **AND** True, and when is a rule with **OR** True? Which one is harder to satisfy?